

First Control Probe

Committee Empowerment in the LLM Naming Game

Operational research note – 29 July 2026

Purpose in one sentence. Keep the naming game of Ashery, Aiello and Baronchelli unchanged, introduce a committee as a controlled experimental intervention, and measure how much knowing the committee’s assigned policy tells us about the population’s future convention.

What are we trying to learn?

Simple statement

A population of fixed-weight LLM agents repeatedly plays the same pairwise naming game used in *Emergent Social Conventions and Collective Bias in LLM Populations*. A small subset of agents is designated as a committee. Across otherwise matched simulation runs, the committee is assigned different convention policies. We ask:

Does changing the committee’s policy reliably change the future of the population, and at what committee size does that influence become informative?

This is different from asking only whether a committed minority sometimes flips consensus. A takeover probability tests one target in one direction. Empowerment treats the committee assignment as the input of a communication channel and the future population state as its output. It therefore measures whether the committee can select among alternative collective futures.

Technical statement

For each independent episode, let G denote a randomized committee policy and let Y_T denote the population outcome at terminal time T . The first quantity is the effective policy empowerment

$$\mathcal{E}_C^{\text{episode}}(T) = I(G; Y_T), \quad (1)$$

where $I(\cdot; \cdot)$ is mutual information. A time-resolved version asks whether the policy predicts a future macrostate after conditioning on the present macrostate:

$$\mathcal{E}_C(\ell) = I(G; Z_{t+\ell} \mid Z_t). \quad (2)$$

The horizon ℓ determines how far into the future the probe looks. These quantities require only logged actions and states. They do **not** require logits, gradients, model weights, or internal chains of thought.

What is inherited from each anchor paper?

Ashery et al.	The exact naming-game mechanics: random pair selection, local coordination payoffs, finite convention pool, rolling agent memory, fixed-weight API calls, and committed agents.
Riedl	The empirical information-theoretic discipline: construct low-dimensional contingency tables, correct finite-sample entropy bias, use time lags, and test against shuffle-based null models. We do <i>not</i> reuse the PID synergy score as the control metric.
Song et al. (EELMA)	The conceptual definition of effective empowerment as conditional mutual information between an intervention/action and a future state. For the structured naming game, direct counting is preferable to a neural InfoNCE estimator.

Scope of this note. This is a diagnostic prototype, not yet a complete theory of controllability. Its purpose is to determine whether an empowerment-style quantity is estimable, nontrivial, robust to null tests, and more informative than the ordinary critical-mass curve.

Game definition and observable variables

The game remains the Science Advances naming game

A trial contains N LLM agents. At each interaction t , two agents (i_t, j_t) are sampled uniformly at random. Each outputs one convention from a finite pool of size W . Matching outputs yield the positive coordination payoff; different outputs yield the negative payoff. Each ordinary agent receives the same prompt structure and a rolling memory of its most recent H interactions, including choices, success or failure, and score. The reference configuration is $N = 24$, $W = 10$, and $H = 5$ for convention-emergence experiments; the committed-minority experiments use $W = 2$ and initialize the population at consensus.

No game rule, prompt logic, memory rule, payoff, partner-sampling rule, or model weight is changed. The additional experimental randomization occurs outside the LLM calls: before an episode, the committee is assigned a policy.

Minimal identifiable intervention

A committee that always promotes one fixed symbol in every run has no variation in its control input. Mutual information cannot identify control from a constant variable. Therefore, at a fixed committee size k , run matched episodes with

$$G \in \{\pi_A, \pi_B, \pi_0\}, \quad (3)$$

where π_A means committee members always output convention A , π_B means they always output B , and π_0 is the no-committee baseline. This does not alter how the game is played; it randomizes which original committed-agent rule is installed in each episode. The cleanest first experiment starts with empty memories and $W = 2$, because the initial-condition distribution can be identical for π_A , π_B , and π_0 . The original consensus-overtaking experiment should be reproduced first as a code-validation test and then used as a second, resilience-oriented experiment.

Variables, in simple and technical language

Symbol	Meaning	Operational definition
N	Population size	Number of LLM agents in one independent simulation episode.
W	Convention-pool size	Number of valid names/symbols available to every agent. Start with $W = 2$.
H	Memory length	Number of past personal interactions included in each ordinary agent's prompt.
T	Episode horizon	Number of pairwise interactions, or the stopping time when consensus is declared.
C	Committee set	Subset of protected/committed agents; $C \subseteq \{1, \dots, N\}$.
$k = C $	Committee size	Number of committee agents; report both k and fraction $\rho = k/N$.
G	Committee policy	Randomized episode-level assignment: always A , always B , or no committee.
$A_{i,t}$	Agent output	Convention emitted by agent i when selected at interaction t .
$M_{i,t}$	Agent memory	Exact rolling history supplied to agent i immediately before its API call.
Z_t	Population macrostate	Low-dimensional summary computed only from observed game logs.
Y_T	Terminal outcome	Final consensus symbol, majority symbol at T , or a three-state outcome $\{A, B, \text{unresolved}\}$.
ℓ	Integration horizon	Number of interactions between the present macrostate and future macrostate.
$I(X; Y)$	Mutual information	Reduction in uncertainty about Y obtained by observing X , measured in bits.

A macrostate requiring no logits

For $W = 2$, define the recent production share of convention A over a window of L interactions:

$$Z_t = \frac{1}{2L} \sum_{s=t-L+1}^t [\mathbf{1}(A_{i_s, s} = A) + \mathbf{1}(A_{j_s, s} = A)]. \quad (4)$$

Here $\mathbf{1}(\cdot)$ is one when the statement is true and zero otherwise. Thus $Z_t = 1$ means all recent outputs were A , $Z_t = 0$ means all were B , and $Z_t \approx 1/2$ indicates a mixed regime. This is a population-level analogue of the production-probability trajectories already plotted in the naming-game paper.

For stable counting, discretize Z_t into two bins in the primary analysis and three bins as a sensitivity test. The exact thresholds must be specified before examining results.

Control quantities and estimation

Quantity 1: terminal effective empowerment

Estimate the channel $p(Y_T | G, k)$ separately for each committee size k . Then compute

$$\mathcal{E}_C^{\text{episode}}(k, T) = \sum_{g, y} p(g, y | k) \log_2 \frac{p(g, y | k)}{p(g | k)p(y | k)}. \quad (5)$$

Interpretation. If the terminal convention is independent of the randomized committee policy, the value is approximately zero bits. If different committee policies reliably produce different terminal conventions, the value increases. With two equiprobable target policies and two perfectly controlled outcomes, the maximum is one bit.

The ordinary critical-mass curve remains a necessary baseline:

$$p_{\text{target}}(k) = \Pr(Y_T = G | G \in \{\pi_A, \pi_B\}, k). \quad (6)$$

Empowerment adds information that this curve alone misses. For example, a population with a strong intrinsic bias toward A may show a high success rate for π_A but a low $I(G; Y_T)$ because changing the committee assignment does not change the outcome.

Quantity 2: time-resolved committee empowerment

Using the macrostate in Eq. (4), estimate

$$\mathcal{E}_C(k, \ell) = I(G; Z_{t+\ell} | Z_t, k). \quad (7)$$

This asks: among episodes currently in comparable population states, does the committee policy distinguish their futures ℓ interactions later? Plotting the value against ℓ separates four behaviours:

- **No influence:** near zero at every horizon.
- **Transient influence:** positive at short horizons, then decays.
- **Delayed amplification:** initially small, then rises as local interactions propagate the intervention.
- **Persistent control:** remains positive at long horizons and predicts the final convention.

Because several time points come from the same episode, episodes—not individual rows—are the independent statistical units. Confidence intervals and permutations must therefore resample or shuffle complete episodes.

Quantity 3: maximum channel capacity (optional)

Classical empowerment maximizes mutual information over the control-input distribution:

$$\mathcal{C}_C(k, T) = \max_{p(g)} I(G; Y_T | k). \quad (8)$$

For the small discrete channel $p(Y_T | G, k)$, compute this with Blahut–Arimoto as a secondary analysis; the balanced experiment remains the primary, directly interpretable estimate.

What exactly is borrowed from Riedl?

Riedl computes time-delayed mutual information and PID from discretized trajectories. The transferable machinery is:

1. **Low-dimensional contingency tables.** Our variables are already discrete; no access to hidden representations is required.
2. **Jeffreys smoothing.** Add 1/2 pseudo-count to every cell before computing probabilities, preventing empty-cell pathologies.
3. **Alternative entropy correction.** Repeat the main result with Miller–Madow correction.
4. **Lag analysis.** Evaluate several pre-registered horizons ℓ rather than selecting the most favourable one.
5. **Shuffle nulls.** Compare observed information against distributions produced by destroying the intervention–future relation while preserving simpler structure.

Required falsification tests

Null test	What it rules out
Episode-label shuffle	Shuffle G across episodes within the same k , model, prompt version, and initial-condition stratum. This destroys control while preserving outcome frequencies.
Temporal shift null	Circularly shift each episode’s Z_t trajectory relative to G or future states. This tests whether apparent information comes only from autocorrelation.
Convention-label swap	Swap $A \leftrightarrow B$ in half the episodes. A valid measure should be invariant to arbitrary symbol names.
No-committee baseline	Confirms that the analysis does not manufacture empowerment from spontaneous symmetry breaking alone.

Do not use EELMA in the first implementation. EELMA is needed when observations and actions are high-dimensional natural-language strings with many paraphrases. Here, the scientific state and action variables are known symbolic quantities. A direct estimator is more transparent and easier to validate. EELMA becomes relevant only after the game is extended to free-form arguments, evidence, or semantic belief states.

Operational experiment and decision criteria

Implementation sequence

1. **Replicate the source game.** Verify the original convergence curves and committed-minority critical-mass result before adding a new metric.
2. **Randomize the probe.** Use $W = 2$, empty initial memories, fixed N, H, T , and assign $G \in \{\pi_A, \pi_B, \pi_0\}$ independently for every episode. Keep model, temperature, prompts, payoffs, and pair sampling fixed.
3. **Sweep committee size.** A first grid is $k \in \{0, 1, 2, 3, 4, 6, 8\}$ for $N = 24$, with equal independent replications for every valid (k, G) cell.
4. **Log complete trajectories.** Store episode ID and seed; model/version and prompt hash; k , committee identities and G ; selected pair and outputs; payoff, scores and memories; Z_t , stopping time and Y_T .
5. **Estimate the channel.** Compute $p(Y_T | G, k)$, terminal MI, lagged conditional MI, episode-bootstrap intervals, and shuffle-null corrected values.
6. **Add richer controls only afterward.** Reasoning prompts, protected memory, verifier access, topology, and adversarial language must each be separately randomized.

Minimum plots

1. Reproduced production trajectories and consensus probability.
2. Channel matrices $p(Y_T = y | G, k)$.
3. $\mathcal{E}_C^{\text{episode}}(k, T)$ versus $\rho = k/N$, with confidence intervals and shuffled-null band.
4. $\mathcal{E}_C(k, \ell)$ versus horizon ℓ for selected k .

5. Target-takeover probability and empowerment on the same committee-size axis.

What would be informative?

The probe is useful only if empowerment separates regimes that final consensus conflates:

- **Critical susceptibility:** empowerment peaks or rises sharply near the critical-mass region because different committee policies select different attractors.
- **Bias without control:** one convention dominates, but MI is low because outcomes do not track committee assignment.
- **Influence without takeover:** short-horizon MI is positive although terminal takeover remains rare.

A null result is also decisive: if MI is merely a monotone rewriting of takeover probability, or cannot be estimated at realistic API budgets, it is unlikely to be the canonical quantity.

Main limitations

- G is fixed within an episode, so this measures influence of a policy assignment, not an optimized adaptive controller.
- The naming game tests convention steering, not truth propagation or reasoning; it is a calibration environment.
- Results may depend on macrostate definition, binning, horizon, and API stochasticity; these require pre-registration and sensitivity analysis.
- MI is unsigned: it measures control, not whether the induced convention is beneficial. Truth and attack control require a later game with external ground truth.

Immediate deliverable after this note. The implementation specification should reproduce the original game, add only episode-level committee-policy randomization, construct the logged variables above, and produce the five diagnostic plots. This is enough to decide whether committee empowerment deserves to anchor the larger agenda.

References. [1] A. F. Ashery, L. M. Aiello, and A. Baronchelli, “Emergent social conventions and collective bias in LLM populations,” *Science Advances*, 2025. [2] C. Riedl, “Emergent Coordination in Multi-Agent Language Models,” *ICLR*, 2026. [3] J. Song, J. Gore, and M. Kleiman-Weiner, “Estimating the Empowerment of Language Model Agents,” arXiv:2509.22504; latest manuscript marked as an ICLR 2026 conference paper.